Sai Soham Venkata Mashetty

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.05
Narayana Junior College	Class XII (Telangana State Board Of Intermediate Education)	2023	96.20%
FIITJEE School	Class X (Board of Secondary Education)	2021	10.00%

SCHOLASTIC ACHIEVEMENTS

Received the **Academic Excellence Award** for outstanding performance in the year 2024-25.**(Issued by IITH)**

Achieved a highest rating of **1479(Specialist)** on Codeforces. (Profile)

Runner up for Creative Design at the **International Youth Robotics Championship 2018** at Thailand

Member of the **NIAS** prodigy program at **IISc Bangalore** from 2017-2019

WORK EXPERIENCE

Oracle

Project Intern, TARS (Ticket Automation Remediation System)

- Built and deployed an end-to-end Agentic AI service to ingest, classify, analyze, remediate, and validate software security findings across large-scale codebases.
- Engineered the orchestration pipeline for evidence persistence, repository/context discovery, multi-repository issue decomposition, repository-scoped work items, autonomous remediation, and ticket resolution.
- Supported Anchore, Parfait, and Fortify , achieving 65% fully autonomous remediation with no manual intervention.

SemXAI

- Built a runtime operator-control layer in C11 that works on the model's intermediate representation (IR), letting users apply reinforcement-learning power-management policies live to a running inference runtime

PROJECTS

Near-Memory Processing Unit for LLM Inference (Link to Project)

C++, GPU accelerators (HBM4 / GDDR7 / LPDDR6), roofline analysis, LLM inference

- Reproduced the AttAcc processing-in-memory simulator and characterised attention and embedding operators with the roofline model, quantifying the bandwidth gap that leaves added compute idle.
- Proposed a half-row-offset, pseudo-channel-level MAC scheduling scheme so one memory domain issues while the other changes rows; timing analysis projects MAC utilisation from 82.8% to 90.6% (+9.4% throughput) at equal average power.

RISC-V Pipelined Processor & Simulator (Link to Project)

C++, RISC-V, Computer Architecture, CMake

- Built a cycle-accurate **5-stage pipelined RISC-V simulator**, implementing forwarding and stall handling.
- Implemented **static, 1/2-bit, Perceptron, and Gshare** branch predictors with cycle-level misprediction analysis.

- Added **automated validation** against Ripes ground truth and benchmarked execution across matrix multiplication, quicksort, binary search, and GCD workloads.

Multi-Client Quiz System with LLM Integration **(Link to Project)**

C++, TCP Sockets, POSIX Threads, Wireshark

- Built a **concurrent TCP client-server** quiz system in C++ supporting simultaneous clients
- Designed an **application-layer protocol** with keep-alive handshaking for session management and dead-peer detection
- Implemented **server-side response caching** for popular quiz genres, reducing median quiz delivery time
- Profiled **8,700+ packets** across **12 Wireshark captures** at **1-10 client loads**, measuring **4.2 ms median request latency** under full concurrency

TECHNICAL SKILLS

Languages: C++, C, Python, SQL, Bash

Systems & Architecture: Linux, x86-64, RISC-V, virtual memory, page tables, POSIX threads, concurrency

Networking: TCP/IP, sockets, protocol design, Wireshark, client-server architecture

Infrastructure & Tooling: Docker, OCI, Git, CMake, CI/CD, MySQL

Runtime & Analysis: ONNX, static analysis (Fortify, Anchore, Parfait), LLM integration, caching

RELEVANT COURSEWORK

Core Courses: Systems for ML, Parallel and Concurrent Computing, Computer Networks, Operating Systems, Data Structures & Algorithms, Computer Architecture, Theory of Computation, Algorithms, Database Management Systems, Software Engineering, Deep Learning, Machine Learning

Math Courses: Probability and Random Variables, Intro to Statistics, Discrete Maths, Linear Algebra, Calculus

EXTRACURRICULARS

Member of the **NSO Lawn Tennis** program at IIT Hyderabad; officiated matches during institute-level tournaments

Completed Certification in **Discover Scuba Diving** program by **PADI**

Rated **1700+ (Expert 2)** in matiks math section